

Chap 3

lecture notes

Date

No.

EOF

- EOF is a constant and its value is -1 in most implementation. This is basically a file marker.
- fscanf reads the contents of a file. It returns -1 (EOF) if it attempts to read the end of the file.

examples

/ copy from indata file to outdata file */*

```
#include <stdio.h>
```

```
void main()
```

```
{ FILE *indata, *outdata;
```

Capital letters

```
char this1; drive name type
```

```
indata = fopen("f:demo33.inf", "r"); rt: string use " " not ' ' open the file for reading only
```

```
outdata = fopen("f:demo33.out", "w"); open the file for writing only
```

```
if (indata == NULL)
```

```
{ printf("Can't open demo33.inf");
```

```
exit(1); The operating system will regard it as a program execution failure and write the data in the file buffer to the disk as part of the clean up process
```

```
if (outdata == NULL)
```

```
{ printf("Can't open demo33.out");
```

```
exit(1);
```

```
while (fscanf(indata, "%c", &this1) != EOF)
```

```
fprintf(outdata, "%c", this1);
```

```
fclose(indata);
```

```
fclose(outdata);
```

```
}
```

pointer

don't forget to close the file

basic syntaxes of fscanf

```
fscanf (stream, format, ...)
```

fscanf is used to read formatted data from a file. It was similar to scanf, but the source of input is different:

- scanf → reads from standard input (keyboard)
- fscanf → reads from a specified file

→ **stream** : File pointer returned by fopen()

format : %d, %5, %f

... : *(no & for strings)* addresses of variables to store ..

Besform

Return value: Returns the number of input items successfully read. Returns EOF (usually -1) or end-of-file or error.

example

/* fscanf & string */

#include <stdio.h>

int main()

{ FILE* fp;

char name[20]; int age;

float score;

fp = fopen("f:data.in", "r");

if (fp == NULL)

{

printf("Cannot open file!");

exit(1);

}

while (fscanf(fp, "%s %d %f", name, &age, &score) == 3)

no need to add &

printf("Name: %s, Age: %d, Score: %.2f\n",

name, age, score);

}

fclose(fp);

return 0;

}

basic syntax of printf

→ same as printf

fprintf(stream, format, ...);

↳ the file pointer

→ the values to be written into the file

Returns the number of items successfully written

subfile

outdata = fopen("f: \\lecture \\ chapter 3 \\ demo33 a.out", "w")